

Week 2 – TA Session

From Detection to Learning

Course: Machine Learning for Signal Processing

By: Amirsadra Khodadadi – Faranak Rezaie

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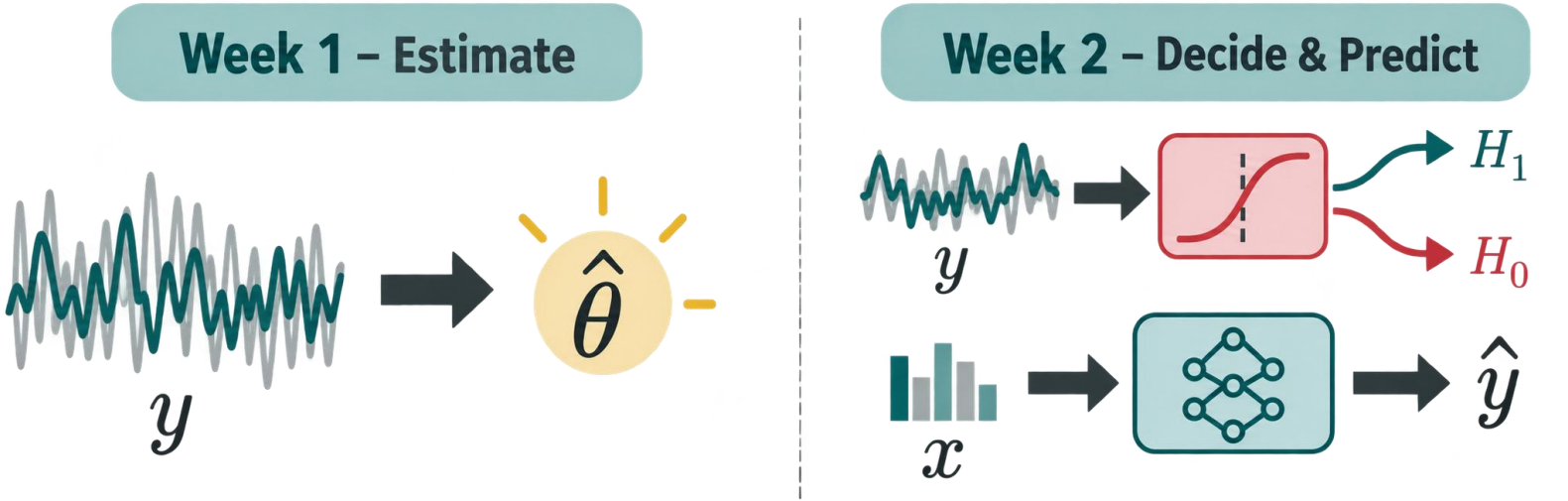
Generalization

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Conclusion

Quick Review: From Estimation to Learning

- *A model may fit the observed data well, but can we trust its decisions and predictions on unseen data?*

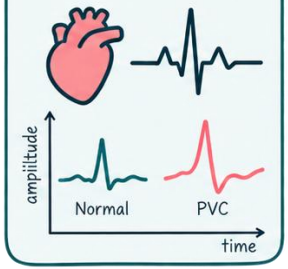


From Detection to Learning, Prediction, and Generalization

We observe noisy data, make decisions, learn predictive models, and ask whether they will work on new data.

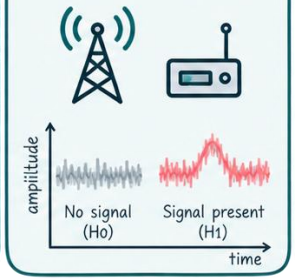
Biomedical Engineering

Classifying ECG beats as normal or PVC.



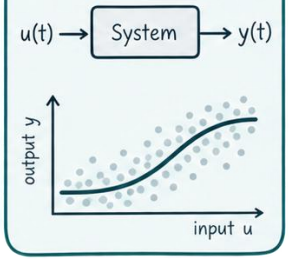
Telecommunications

Detecting whether a weak known signal is present in noise.



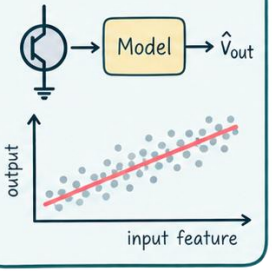
Control & Sensors

Learning a predictive relationship from noisy measurements.



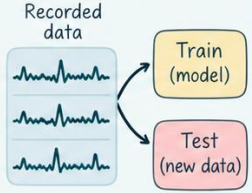
Electronics

Predicting an output from measured input features.



Signal Processing

Testing whether a learned model truly generalizes to new recordings.

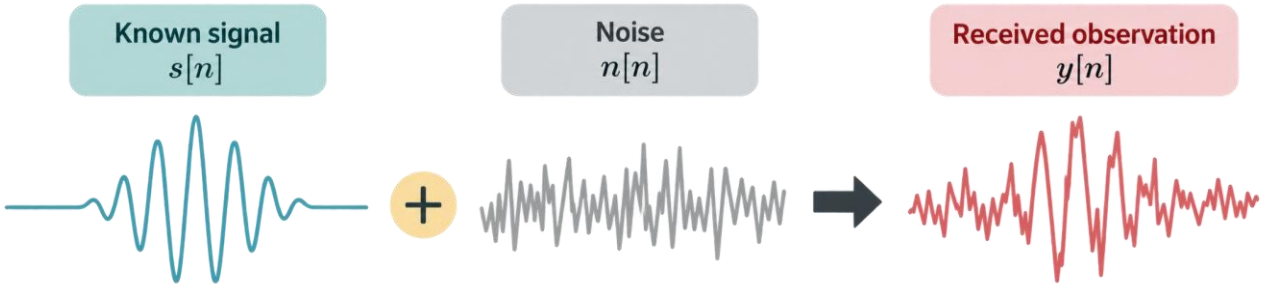


How can we learn reliable decisions and predictions from noisy data — and know they generalize?

01

Statistical Detection

Can We Detect a Signal Hidden in Noise?



Observation Model

$$H_0 : \mathbf{y} = \mathbf{n}$$
$$H_1 : \mathbf{y} = A\mathbf{s} + \mathbf{n}$$

- $(\mathbf{s})$:Known signal waveform
- (A):Known positive amplitude
- $(\mathbf{n})$:Additive Gaussian noise
- $(\mathbf{y})$:Received observation

Goal

Decide whether the signal is present while controlling the probability of false alarms.

Q1:

Suppose we lower the detection threshold to detect weaker signals.
What happens to:
The probability of detecting a signal?
The probability of generating a false alarm?
Can we improve one without affecting the other?

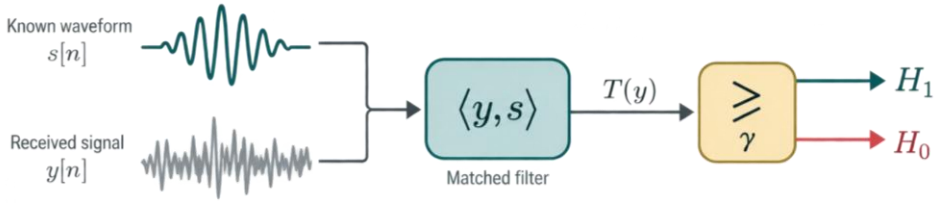
From a Waveform to a Decision

For a known signal + white Gaussian noise:

$$[T(\mathbf{y}) = \mathbf{s}^T\mathbf{y}]$$

Decision rule (γ: threshold):

$$\left[T(\mathbf{y}) \underset{H_0}{\overset{H_1}{\gtrless}} \gamma\right]$$





- The detection score measures evidence for the signal.
- The threshold determines when we declare detection.
- Changing the threshold changes the operating point.

EXPERIMENT 1

Detection Summary

Thresholds create a trade-off

Lower thresholds generally increase both detection probability and false-alarm probability.

ROC: performance across thresholds

A ROC curve separates the quality of the detection score from a particular threshold choice.

SNR affects detectability

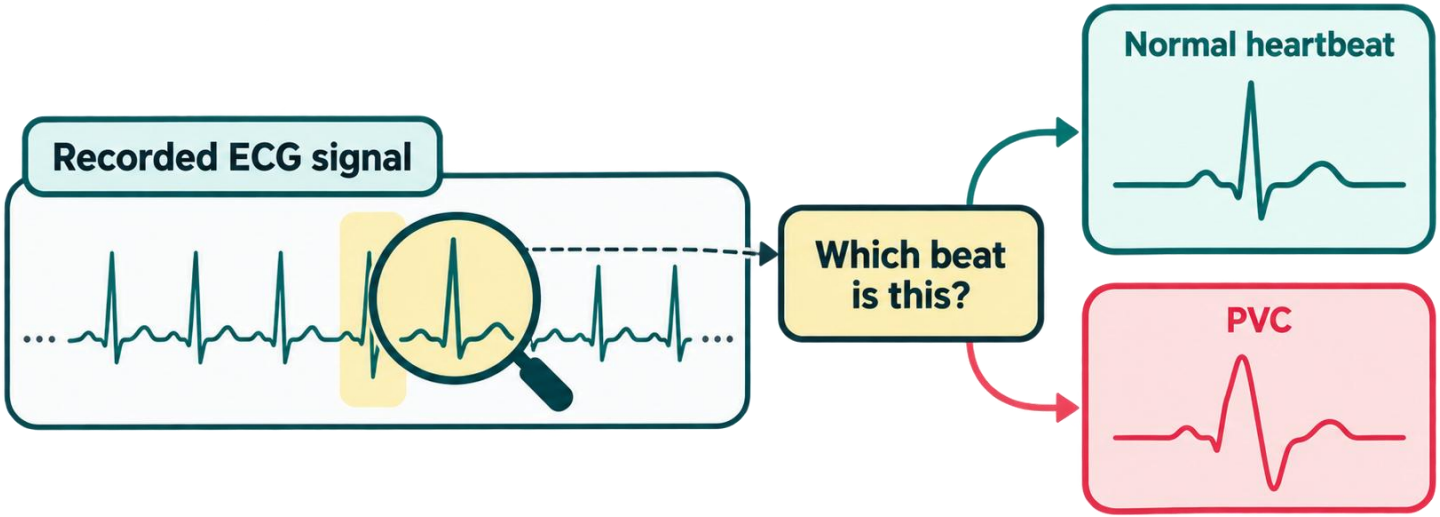
Higher SNR makes the hypotheses easier to distinguish in Gaussian signal model.

02

Learning to Classify

What If the Statistical Model Is Unknown?

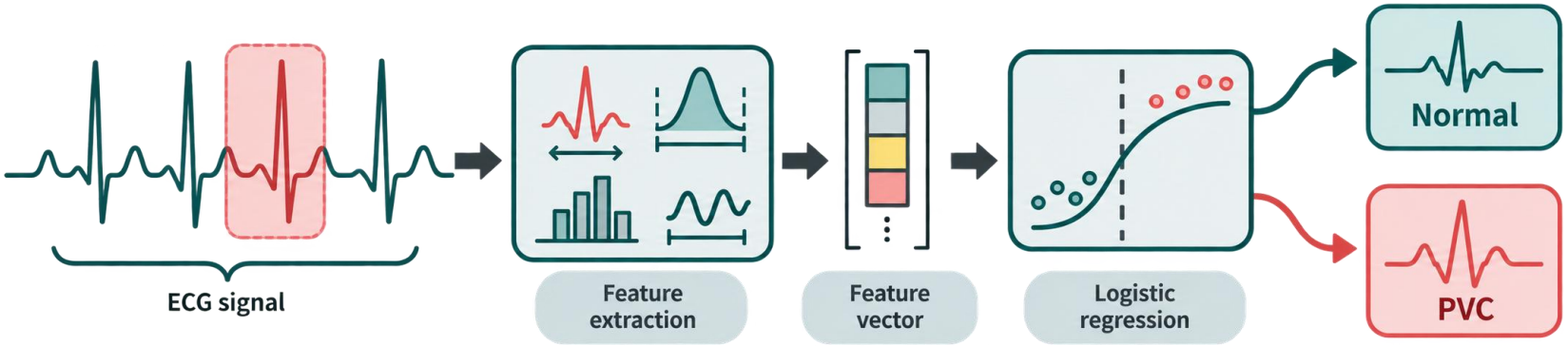
Example: ECG Beat Classification



Dataset: MIT-BIH Arrhythmia Database (PhysioNet)

From Raw Signals to Machine Learning Features

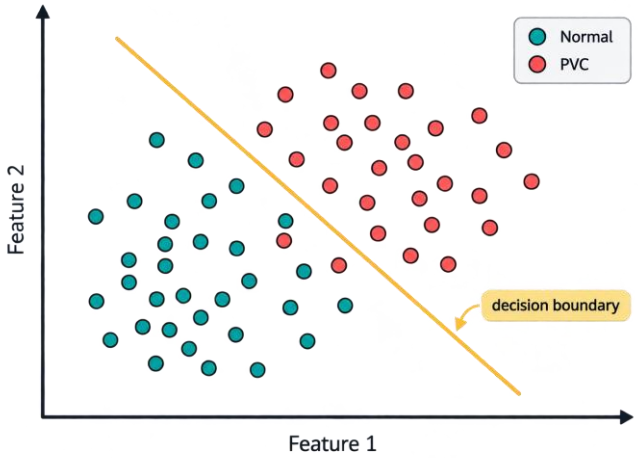
The classifier can only learn from the information represented by its input features.

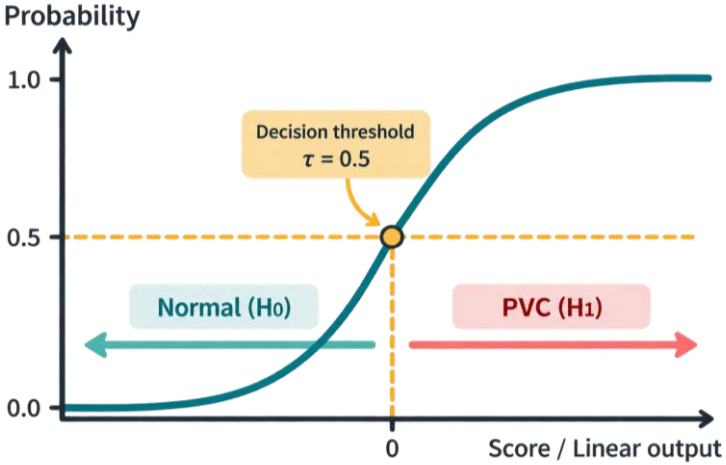


Examples of ECG features:

- Signal amplitude
- Local signal energy
- Time between heartbeats

Learning a Decision Rule





Logistic Regression

Learn a model from labeled examples.

$$\sigma\left(\mathbf{w}^T\mathbf{z} + b\right)$$

Probability to decision

Predict H_1: $\hat{H} = H_1, \quad P(H_1|\mathbf{z}) > \tau$

Predict H_0: $\hat{H} = H_0, \quad P(H_1|\mathbf{z}) \leq \tau$

Q2:

Consider a rare-event detection system, where out of 10,000 observations, only 1% contain an event. Suppose our classifier achieves:

$$[P_D = 0.80]$$
$$[P_{FA} = 0.01]$$

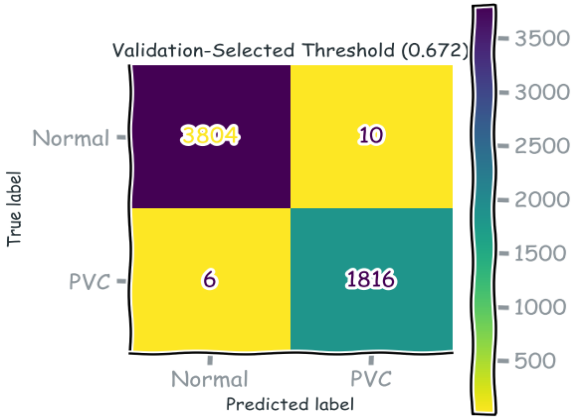
When the classifier predicts that an event is present, what percentage of those predictions are actually correct? Would you consider this a reliable detector?

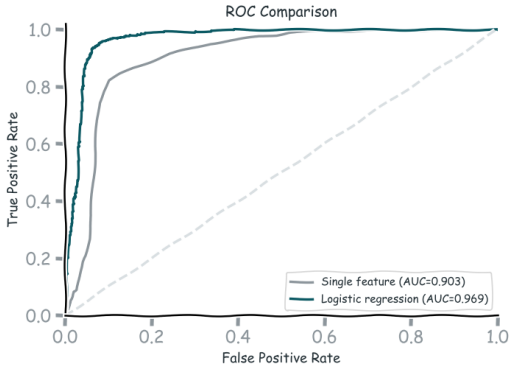
EXPERIMENT 2

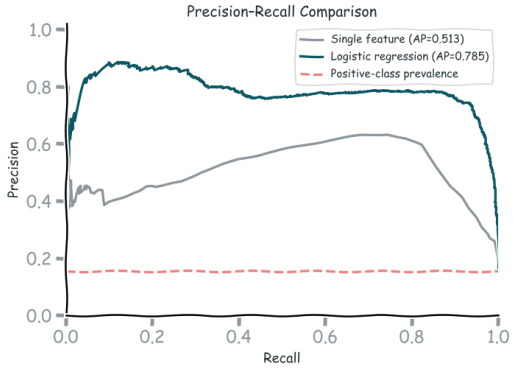
Accuracy Is Not the Whole Story

Different metrics answer different questions

- **Accuracy** → How frequently are our predictions correct overall?
- **Recall / Detection Probability** → How many actual positive events are detected?
- **Precision** → How many predicted positive events are truly positive?





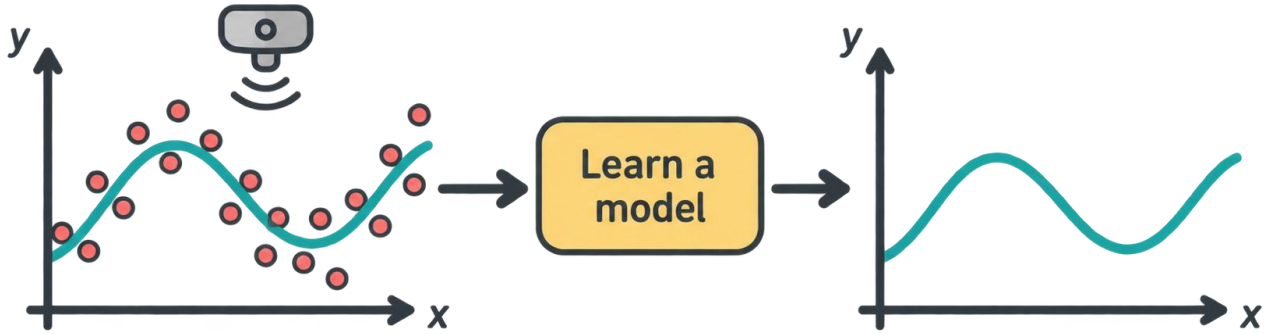


03

Regression & Regularization

Can We Learn the Relationship Behind Noisy Data?

Example: a sensor measures a quantity whose relationship with the input is nonlinear.



Observation Model

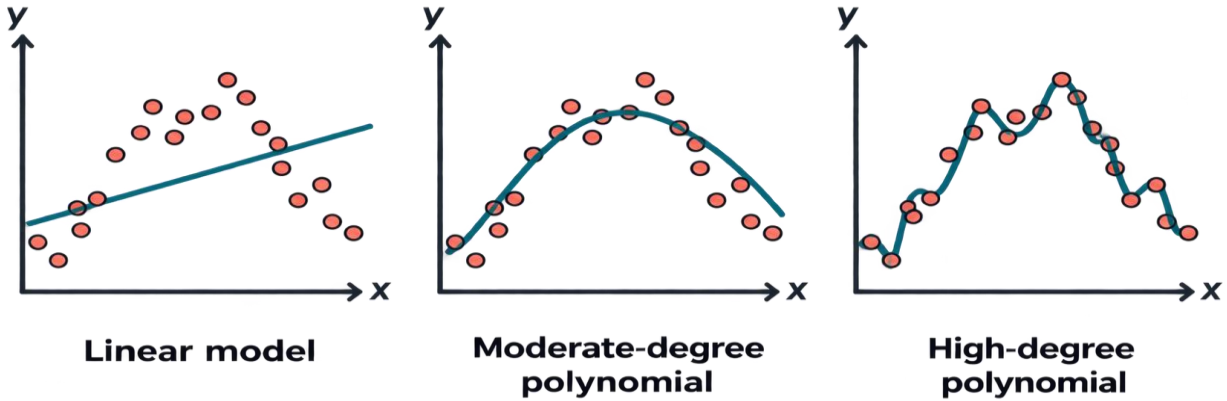
$$y_i = f(x_i) + \epsilon_i$$

Goal

Learn a function that predicts the underlying relationship from noisy observations.

Linear Regression Is Not Always a Straight Line

Nonlinear model: $\beta_0 + \beta_1 x + \beta_2 x^2 + \cdots + \beta_p x^p \rightarrow$ linear in the parameters ($\boldsymbol{\beta}$).



Increasing model complexity

- Low complexity may miss important patterns.
- Greater flexibility can improve the training fit.
- Excessive flexibility can make predictions sensitive to the training data.

Q3:

Suppose we increase the degree of a polynomial regression model from 1 to 12. The training error continues to decrease. Does this guarantee that the model will perform better on unseen data? If not, how should we determine the appropriate model complexity?

Can Regularization Improve Prediction?

Regularization:

$$\arg\min_{\boldsymbol{\beta}}\left[|\mathbf{y}-\mathbf{X}\boldsymbol{\beta}|_2^2+\lambda\Omega(\boldsymbol{\beta})\right]$$

Ridge Regression	**LASSO**
$\lvert\boldsymbol{\beta}\rvert_2^2$	$\lvert\boldsymbol{\beta}\rvert_1$
Encourages smaller coefficients.	Can produce sparse solutions.

EXPERIMENT 3

A Better Fit Is Not Always a Better Model

Training performance

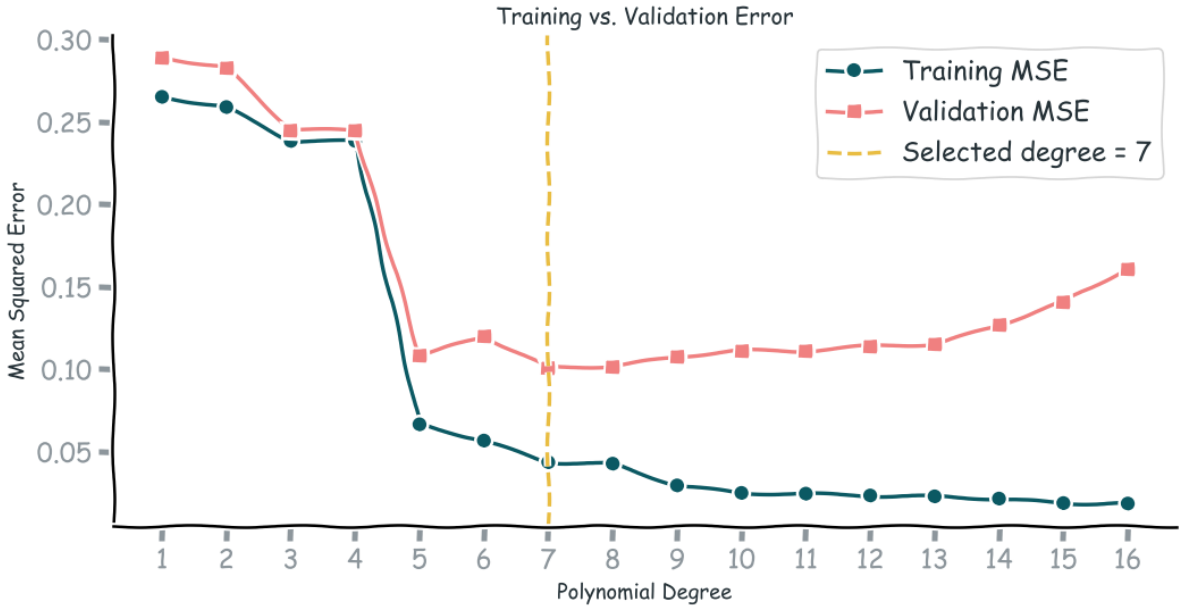
- Measures how well the model fits the observations used for learning.

Validation performance

- Helps us select model complexity and regularization strength.

Test performance

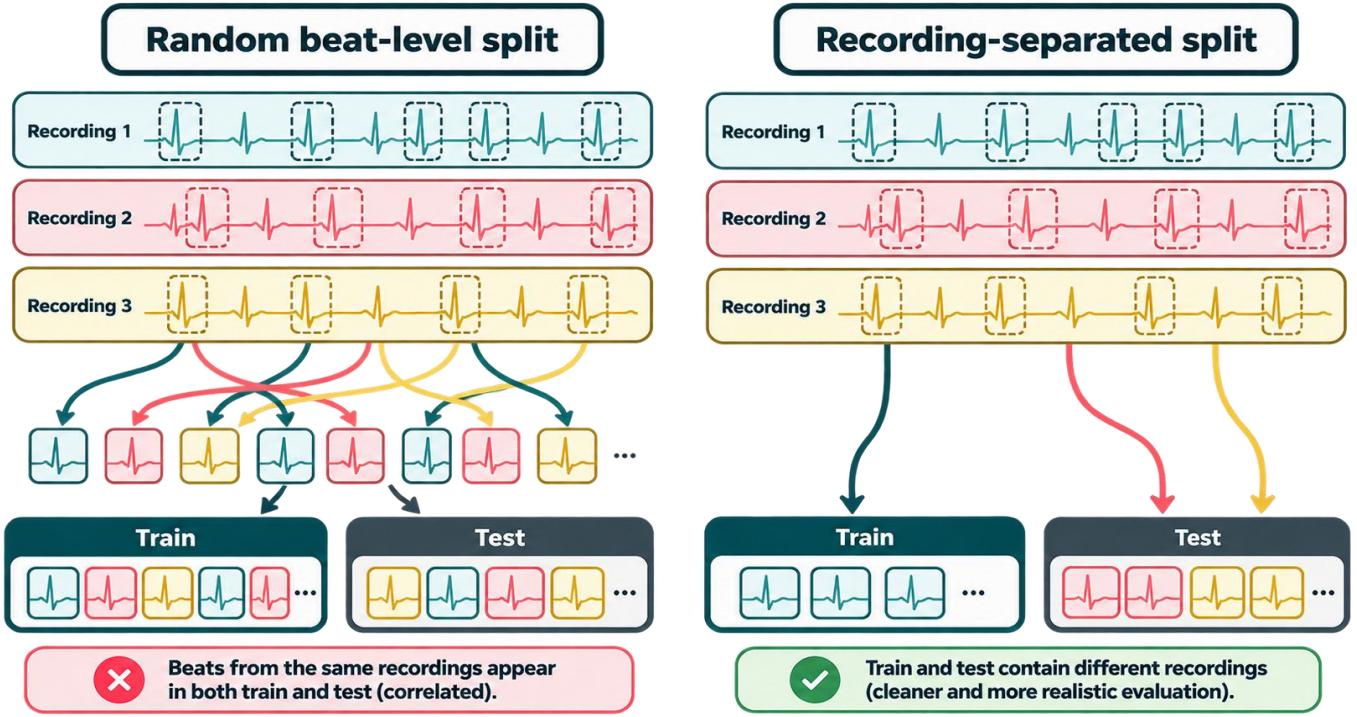
- Estimates how well the selected model performs on unseen observations from the target evaluation setting.


Training vs. Validation Error
Training MSE
Validation MSE
Selected degree = 7
Mean Squared Error
0.30
0.25
0.20
0.15
0.10
0.05
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
Polynomial Degree

04

Generalization

Are Our Training and Test Samples Independent?



Q4:

Suppose an ECG classifier achieves excellent test accuracy when heartbeats are randomly divided into training and test sets. However, both sets contain samples from the same patients. Can we conclude that the classifier will perform equally well on completely new patients? What would be a more appropriate evaluation strategy?

EXPERIMENT 4

05

Conclusion

Final Comparison

Problem	Main Question	Takeaway
Statistical detection	Is the signal present?	A decision threshold controls the detection trade-off.
Classification	Can we learn a decision rule?	Features, training objectives, and evaluation metrics matter.
Regression	Can we predict a continuous quantity?	Better training fit does not guarantee better prediction.
Generalization	Can we trust performance on unseen data?	Evaluation must reflect the intended application.

THANKS!

Do you have any questions?

amirsadra8090@gmail.com
faranak.rezaie81@gmail.com